

Find the intersection of two sorted arrays.

OR in other words,

Given 2 sorted arrays, find all the elements which occur in both the arrays.

Input Format

- The first line contains T, the number of test cases. Following T lines contain:
  1. Line 1 contains N1, followed by N1 integers of the first array
  2. Line 2 contains N2, followed by N2 integers of the second array

Output Format

The intersection of the arrays in a single line

Example

Input:

```
1
3 10 17 57
6 2 7 10 15 57 246
```

Output:

```
10 57
```

Input:

```
1
6 1 2 3 4 5 6
2 1 6
```

Output:

```
1 6
```

**For example:**

| Input                                    | Result |
|------------------------------------------|--------|
| 1<br>3 10 17 57<br>6<br>2 7 10 15 57 246 | 10 57  |

```
1 #include <stdio.h>
2
3 int main()
4 {
5     int T;
6     scanf("%d", &T);
7
8     while (T--)
9     {
10         int n1, n2;
11         scanf("%d", &n1);
12
13         int a[n1];
14
15         for (int i = 0; i < n1; i++)
16             scanf("%d", &a[i]);
17
18         scanf("%d", &n2);
19
20         int b[n2];
21
22         for (int i = 0; i < n2; i++)
23             scanf("%d", &b[i]);
24
25         int i = 0, j = 0;
26
27         while (i < n1 && j < n2)
28         {
29             if (a[i] == b[j])
30             {
31                 printf("%d", a[i]);
32                 i++;
33                 j++;
34
35                 if (i < n1 && j < n2)
36                     printf(" ");
37             }
38             else if (a[i] < b[j])
39             {
40                 i++;
41             }
42             else
43             {
44                 j++;
45             }
46         }
47
48         printf("\n");
49     }
50
51     return 0;
52 }
```

|   | Input                                                | Expected | Got   |   |
|---|------------------------------------------------------|----------|-------|---|
| ✓ | 1<br><br>3 10 17 57<br><br>6<br><br>2 7 10 15 57 246 | 10 57    | 10 57 | ✓ |
| ✓ | 1<br><br>6 1 2 3 4 5 6<br><br>2<br><br>1 6           | 1 6      | 1 6   | ✓ |

Passed all tests! ✓